

핵심 : Matrix에서의 PCA(SVD)를 텐서에서의 Tucker decomposition(HOSVD)로 확장.

- Singular value decomposition (SVD) -

In general matrix $A \in \mathbb{R}^{m \times n}$ can be decomposed,

$$A = U \Sigma V^T$$

(where $U \in \mathbb{R}^{m \times m}$: left singular vectors,
 $\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_{\min(m,n)} \\ & & & & 0 \end{bmatrix} = \Sigma \in \mathbb{R}^{m \times n}$: singular values,
 $V \in \mathbb{R}^{n \times n}$: right singular vectors)

$$\Rightarrow A^T A = V \Sigma^T U^T U \Sigma V^T = V (\Sigma^T \Sigma) V^T \\ = \sigma^2 V V^T$$

$$\Rightarrow A^T A V = \sigma^2 V$$

$\Rightarrow \sigma^2$ is eigenvalue, V is eigenvector of $A^T A$
like wise,

U is eigenvector of $A A^T$

U, V 는 각각 column wise orthonormal 이고 회전 또는 대칭 이동을 포함.
 Σ 는 diagonal 이고 크기 변환을 포함.

동일한 구조가 Tucker decomposition 에도 적용 됨.

- Tucker decomposition -

The Tucker decomposition is a form of a higher-order PCA.

It decomposes a tensor into a core tensor multiplied (or transformed) by a matrix along each mode.

in the three-way case, where $\mathbb{X} \in \mathbb{R}^{I \times J \times K}$

$$\mathbb{X} \approx \underbrace{\mathbb{G}}_{= \mathbb{A} \circ \mathbb{B} \circ \mathbb{C}} \times_1 \mathbb{A} \times_2 \mathbb{B} \times_3 \mathbb{C} = \sum_{p=1}^P \sum_{q=1}^Q \sum_{r=1}^R g_{pqr} a_p \circ b_q \circ c_r = [[\mathbb{G}; \mathbb{A}, \mathbb{B}, \mathbb{C}]]$$

(where, $\mathbb{A} \in \mathbb{R}^{I \times P}$, $\mathbb{B} \in \mathbb{R}^{J \times Q}$, $\mathbb{C} \in \mathbb{R}^{K \times R}$ are the factor matrices

$\mathbb{G} \in \mathbb{R}^{P \times Q \times R}$ is called the core tensor

' $\times_n$ ' is n-mode product)

Elementwise,

$\hookrightarrow \mathbb{G}$ Tensor is n-mode all Matrix ($\mathbb{A}, \mathbb{B}, \mathbb{C}$) $\frac{1}{2} \frac{1}{2} \frac{1}{2}$).

$$x_{ijk} \approx \sum_{p=1}^P \sum_{q=1}^Q \sum_{r=1}^R g_{pqr} a_{ip} b_{jq} c_{kr} \quad \text{for } i=1, \dots, I, j=1, \dots, J, k=1, \dots, K.$$

If P, Q, R are smaller than I, J, K , the core tensor $\mathbb{G}$ can be thought of as a compressed version of origin tensor ($\mathbb{X}$).

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In fact, CP can be viewed as a special case of Tucker where the core tensor is superdiagonal ($p=q=r$) and $P=Q=R$.

$$\sum_p \sum_q \sum_r g_{pqr} a_p \circ b_q \circ c_r \xrightarrow{p=q=r} \sum_r \lambda_r a_r \circ b_r \circ c_r$$

The Flattening forms of Tucker decomposition are

$$X_{(1)} \approx A G_{(1)} (C \otimes B)^T$$

$$X_{(2)} \approx B G_{(2)} (C \otimes A)^T$$

$$X_{(3)} \approx C G_{(3)} (B \otimes A)^T$$

- Tucker 1 decomposition -

The Tuck 1 decomposition sets two of the factor matrices to be the identity matrix.

e.g. $B = C = I$

$$\Rightarrow X = G X_1 A X_2 I X_3 I$$

$$= G X_1 A = [[G; A, I, I]]$$

$$\therefore X_{(1)} = A G_{(1)}$$

- Tucker 2 decomposition -

Tucker 2 decomposition of a third-order array sets one of the factor matrices to be the identity matrix.

e.g. $C = I$

$$\Rightarrow X = G X_1 A X_2 B X_3 I = G X_1 A X_2 B = [[G; A, B, I]]$$

변환이 필요 없는

mode에 적용

- The n-Rank -

The n-Rank of $\mathbb{X}$, denoted $\text{rank}_n(\mathbb{X})$, is the column rank of $\mathbb{X}_{(n)}$. If we let $R_n = \text{rank}_n(\mathbb{X})$ for $n = 1, \dots, N$, then we can say that $\mathbb{X}$ is a rank- $(R_1, R_2, \dots, R_N)$ tensor.

Trivially, $R_n \leq I_n$ for all $n = 1, \dots, N$.

- Higher order SVD (HOSVD) -

The basic idea is to find those components that best capture the variation in mode n , independent of the other modes. This is sometimes referred to as the "Tucker I" method.

When $R_n < \text{rank}_n(\mathbb{X})$ for one or more n , the decomposition is called the truncated HOSVD. In fact, the core tensor of the HOSVD is all-orthogonal, which has relevance to truncating the decomposition. The truncated HOSVD is not optimal in terms of giving the best fit as measured by the norm of the difference.

def HOSVD($\mathbb{X}, R_1, \dots, R_N$):
for n in range(N):
 $A^{(n+1)} \leftarrow R_n$ leading left singular vectors of $\mathbb{X}_{(n+1)}$
 $G = \mathbb{X} \times_1 A^{(1)T} \dots \times_N A^{(N)T}$
return $G, A^{(1)}, \dots, A^{(N)}$

$R_n < \text{Rank}_n(\mathbb{X})$ 일 때 손실 발생.

- higher-order orthogonal iteration (HOOI)-

If we assume that $\mathbb{X}$ is a tensor of size $I_1 \times I_2 \times \dots \times I_N$, then the optimization problem that we wish to solve is

$$\min_{G, A^{(1)}, \dots, A^{(N)}} \|\mathbb{X} - [G; A^{(1)}, A^{(2)}, \dots, A^{(N)}]\|^2$$

$$\left(\text{where, } G \in \mathbb{R}^{R_1 \times R_2 \times \dots \times R_N}, \right.$$

$$A^{(n)} \in \mathbb{R}^{I_n \times R_n} \text{ and columnwise orthogonal for } n = 1, \dots, N \left. \right)$$

$$\begin{aligned} \Rightarrow & \|\mathbb{X} - [G; A^{(1)}, A^{(2)}, \dots, A^{(N)}]\|^2 \\ &= \|\mathbb{X}\|^2 - 2\langle \mathbb{X}, [G; A^{(1)}, \dots, A^{(N)}] \rangle + \|[G; A^{(1)}, \dots, A^{(N)}]\|^2 \\ &= \|\mathbb{X}\|^2 - 2\langle \mathbb{X} \times_1 A^{(1)\top} \times_2 A^{(2)\top} \dots \times_N A^{(N)\top}, G \rangle + \|G\|^2 \\ &\quad (\because A^{(n)} \text{ is column wise } \cancel{\text{orthogonal}} \text{ orthonormal}). \\ &= \|\mathbb{X}\|^2 - 2\langle G, G \rangle + \|G\|^2 = \|\mathbb{X}\|^2 - \|G\|^2 \\ &= \|\mathbb{X}\|^2 - \|\mathbb{X} \times_1 A^{(1)\top} \times_2 A^{(2)\top} \dots \times_N A^{(N)\top}\|^2 \end{aligned}$$

Since, $\|\mathbb{X}\|$ is constant,

we should maximize $\|\mathbb{X} \times_1 A^{(1)\top} \times_2 A^{(2)\top} \dots \times_N A^{(N)\top}\|$

$$\Rightarrow \max_{A^{(n)}} \|\mathbb{X} \times_1 A^{(1)\top} \times_2 A^{(2)\top} \dots \times_N A^{(N)\top}\|$$

$$\Rightarrow \max_{A^{(n)}} \|A^{(n)\top} W\| \quad (\text{where, } W = \mathbb{X}_{(n)} (A^{(1)} \otimes \dots \otimes A^{(n-1)} \otimes A^{(n+1)} \otimes \dots \otimes A^{(N)})$$

The solution can be determined using SVD, simply set $A^{(n)}$ to be the R_n leading left singular vectors of W .

Tucker decomposition : 수학적 모델

$\Rightarrow$ $\left\{ \begin{array}{l} \text{HOSVD : 각 mode의 flattening Matrix에 SVD를 적용.} \\ \text{반복은 없고, 최적으로 근사를 보장 못함.} \\ \text{HOOI : HOSVD의 값을 초기값으로 시작하여, 수렴할 때까지} \\ \text{반복 근사하는 알고리즘. 최적 해는 아닐 수 있음.} \end{array} \right.$